

Cross-target attribution consistency.

To check whether the model's attributions are internally coherent, we compare, per instruction, how strongly each PTX instruction is attributed to the slowdown prediction versus to each PMU-based prediction (under both isolation and contention).

Averaging this comparison across a sample of profiled kernels produces a heatmap of pairwise similarity between targets. The pattern is structured and physically interpretable.

Under contention, the instructions identified as responsible for the slowdown prediction closely match those identified as responsible for the warp-occupancy and L2 read-hit-rate predictions, which is consistent with the underlying hardware mechanism, since memory stalls simultaneously reduce warp issue rate and cache hit rate.

The same instructions show a clear negative relationship with short-scoreboard-stall attributions, which reflect register-dependency stalls rather than memory stalls (a distinct, opposing mechanism from contention-driven slowdown). This anti-correlation is itself a physically meaningful distinction that the model recovered directly from PTX code, without being given any explicit notion of stall types. Together, these results suggest that the model's attributions reflect a coherent, mechanistically grounded representation of contention behavior, rather than independently fitting each target.

Avg instruction cosine similarity across 42 kernels

